13 Diagram 1 shows a ripple tank experiment in which plane waves are diffracted through a narrow slit in a metal sheet.

Diagram 2 shows the same tank with a slit of greater width.

In each case, the pattern of the waves incident on the slit and the emergent pattern are shown.

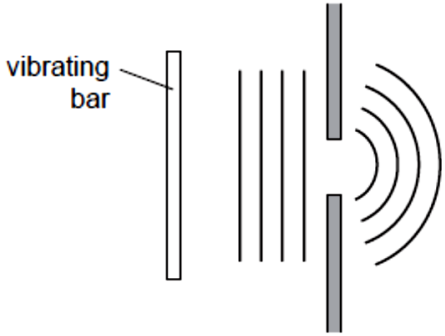

diagram 1

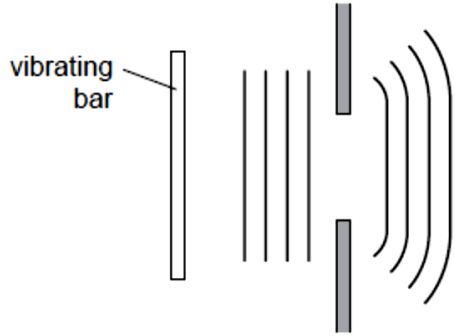

diagram 2

Which action would cause the waves in diagram 1 to produce an emergent pattern closer to that shown in diagram 2?

A Increasing the frequency of vibration of the bar.

B Increasing the speed of the waves by making the water in the tank deeper.

C Reducing the amplitude of vibration of the bar.

D Reducing the length of the vibrating bar.